play a role in the maintenance of fecal – oral route of contamination.

There are two primary types of food related diseases: food – borne infections and food intoxications or food poisoning.

HOTS

Name some of the uncooked foods that have been implicated in food borne disease transmission?

5.3.1 Food Borne Infection

Food borne infection involves the ingestion of the pathogen followed by growth in the host, including tissue invasion and/or the release of toxins. The major diseases of this type are summarized in table (5.1).

5.3.2 Food Poisoning

Food borne intoxication (or) food poisonings is caused by ingesting food containing toxins formed by bacteria which resulted from the bacterial growth in the food item. Food poisoning refers to the toxicity introduced into food by microorganisms and their products.

Microbial growth in food products also can results in food intoxication. Intoxication produces symptoms shortly after the food is consumed because growth of the disease – causing microorganism is not required. Toxins produced in the food can be associated with microbial cells or can be released from the cells.

Food poisoning is caused by various factors as follows.

Table 5.1: Major Food – Borne Infectious Diseases

S.No	Disease	Organism	Incubation period and characteristics	Major Foods Involved
1.	*Salmonellosis*	*Salmonella typhimurium* *S. enteritidis*	8–48 hr Enterotoxin and cytotoxins	Meat, poultry, Fish, eggs, dairy product.
2.	*Campylobacteriosis*	*Campylobacter jejuni*	Usually 2–10. Most toxin heat labile	Milk, or, poultry product, water.
3.	*Listeriosis*	*L. monocytogenes*	Varying periods. Related to meningitis and abortion newborns and the elderly.	Meat products, especially pork and milk.
4.	*Escherichia coli* (Diarrhea and colitis)	*Escherichia coli*, includes serotype 0157:H7	24–72 hrs Entero toxigenic Positive and negative strains : hemorrhagic colitis	Cooked ground beef, raw milk
5.	*Shigellosis*	*Shigella sonnei*, S. *flexneri*	24–72 hrs	Egg products, puddings
6.	*Vibrio parahamolyticus*	*Vibrio parahaemolyticus*	16–48 hr	Seafood, shellfish

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